

B2: Symmetry and Relativity

Model Solutions — Trinity Term 2015

Conventions. Metric $\eta_{\mu\nu} = \text{diag}(+, -, -, -)$. $u^\mu = (\gamma c, \gamma \mathbf{v})$, $u_\mu u^\mu = c^2$. $p^\mu = (E/c, \mathbf{p})$. SI units; $c = 2.998 \times 10^8 \text{ m s}^{-1}$, $m_e c^2 = 0.511 \text{ MeV}$.

Question 1 — Synchrotron radiation: aberration, proper acceleration, storage ring

Problem. Aberration formula for photon emission angle; show 50% of isotropic emission lies in cone $\theta < \cos^{-1}(v/c)$. For $\mathbf{a} \perp \mathbf{v}$, $a_0 = \gamma^2 a$. Storage ring $R = 561 \text{ m}$, $E = 3 \text{ GeV}$ electrons: (i) energy loss per turn, (ii) angular divergence, (iii) pulse length, (iv) frequency spread $\propto \gamma^3$, (v) spectral region.

(a) Aberration of a photon

Take $\mathbf{v} = v\hat{\mathbf{x}}$, lab angle θ , rest-frame angle θ' . Photon 4-momentum $p^\mu = (E/c)(1, \cos \theta, \sin \theta, 0)$. Boost from lab to rest frame ($\beta = v/c$):

$$E'/c = \gamma(E/c - \beta E \cos \theta/c).$$

Angular component:

$$(E'/c) \cos \theta' = \gamma(E \cos \theta/c - \beta E/c).$$

Divide:

$$\cos \theta' = \frac{\cos \theta - \beta}{1 - \beta \cos \theta}. \quad (1)$$

Invert (boost $-\beta$):

$$\boxed{\cos \theta = \frac{\cos \theta' + \beta}{1 + \beta \cos \theta'}.} \quad (2)$$

Isotropic in rest frame: half the photons have $\cos \theta' > 0$. Map $\cos \theta' = 0$ to the lab via (2):

$$\cos \theta = \beta = v/c.$$

Hence half the photons fall inside the lab cone:

$$\boxed{\theta < \cos^{-1}(v/c).} \quad (3)$$

(b) Proper acceleration for $\mathbf{a} \perp \mathbf{v}$

4-velocity $u^\mu = (\gamma c, \gamma \mathbf{v})$. Differentiate:

$$\frac{du^\mu}{d\tau} = \gamma \frac{d}{dt}(\gamma c, \gamma \mathbf{v}).$$

Use $d\gamma/dt = \gamma^3(\mathbf{v} \cdot \mathbf{a})/c^2$; here $\mathbf{v} \cdot \mathbf{a} = 0$:

$$d\gamma/dt = 0.$$

Therefore:

$$\frac{du^\mu}{d\tau} = \gamma(0, \gamma \mathbf{a}) = (0, \gamma^2 \mathbf{a}).$$

The proper acceleration is the magnitude of the spatial 4-acceleration in the rest frame; here:

$$\boxed{a_0 = \gamma^2 a.} \quad (4)$$

(i) Radiative loss per revolution

Lorentz factor:

$$\gamma = \frac{E}{m_e c^2} = \frac{3 \times 10^9 \text{ eV}}{0.511 \times 10^6 \text{ eV}} = 5871.$$

Larmor + boost via (4): $a_0 = \gamma^2 c^2/R$, so radiated power

$$P = \frac{\mu_0 q^2 \gamma^4 c^3}{6\pi R^2}.$$

Time per revolution at $v \simeq c$:

$$T = 2\pi R/c.$$

Loss per turn:

$$U_0 = PT = \frac{\mu_0 q^2 \gamma^4 c^2}{3R}.$$

Rewrite via $\mu_0 c^2 = 1/\epsilon_0$:

$$\boxed{U_0 = \frac{q^2 \gamma^4}{3\epsilon_0 R}.} \quad (5)$$

Numerator $q^2 \gamma^4$:

$$(1.602 \times 10^{-19} \text{ C})^2 (5871)^4 = 2.565 \times 10^{-38} \text{ C}^2 \cdot 1.188 \times 10^{15} = 3.048 \times 10^{-23} \text{ C}^2.$$

Divide by $3\epsilon_0 R$:

$$3\epsilon_0 R = 3(8.854 \times 10^{-12})(561 \text{ m}) = 1.490 \times 10^{-8} \text{ F m}^{-1} \cdot \text{m}.$$

Quotient:

$$U_0 = 2.046 \times 10^{-15} \text{ J} = 12.77 \text{ keV}.$$

$$\boxed{U_0 \approx 12.8 \text{ keV per electron per turn.}} \quad (6)$$

(ii) Angular divergence

Cone half-angle from (2) with $\cos \theta' = 0$ at $\beta \rightarrow 1$:

$$1 - \cos \theta \approx \theta^2/2 = 1 - \beta \approx 1/(2\gamma^2).$$

Hence:

$$\boxed{\Delta\theta \sim 1/\gamma = 1.7 \times 10^{-4} \text{ rad} \approx 170 \text{ } \mu\text{rad.}} \quad (7)$$

(iii) Received pulse length

Beam sweeps the detector while the electron's velocity vector turns through $\sim 1/\gamma$. Arc length swept:

$$\ell = R/\gamma.$$

Emission time interval:

$$\Delta t_e = \ell/v \approx R/(\gamma c).$$

Photons from the trailing end of the arc nearly catch the leading photon; received interval shortens by $(1 - v/c) \approx 1/(2\gamma^2)$:

$$\Delta t_r = \Delta t_e(1 - v/c) \approx \frac{R}{2\gamma^3 c}.$$

Numerical: $2\gamma^3 c = 2(5871)^3(3 \times 10^8 \text{ m s}^{-1})$:

$$2\gamma^3 c = 1.214 \times 10^{23} \text{ m s}^{-1}.$$

Divide:

$$\Delta t_r = 561 \text{ m} / 1.214 \times 10^{23} \text{ m s}^{-1} = 4.6 \times 10^{-21} \text{ s}.$$

Including the prefactor commonly taken without the $1/2$ ($\Delta t_r \sim R/(\gamma^3 c)$) one finds ~ 9 as.

$$\boxed{\Delta t_r \sim \frac{R}{\gamma^3 c} \approx 9 \text{ as.}} \quad (8)$$

(iv) $\Delta\omega \propto \gamma^3$

Frequency spread $\sim 1/\Delta t_r$:

$$\Delta\omega \sim 2\pi/\Delta t_r.$$

Insert (8):

$$\boxed{\Delta\omega \sim \frac{\gamma^3 c}{R} \propto \gamma^3.} \quad (9)$$

(v) Spectral region

Critical photon energy:

$$\hbar\omega_c \sim \hbar\gamma^3 c/R.$$

Compute $\gamma^3 c/R$:

$$\gamma^3 c/R = (5871)^3(3 \times 10^8 \text{ m/s})/561 \text{ m} = 1.08 \times 10^{17} \text{ s}^{-1}.$$

Energy:

$$\hbar\omega_c \sim (1.055 \times 10^{-34} \text{ J s})(1.08 \times 10^{17} \text{ s}^{-1}) = 1.14 \times 10^{-17} \text{ J} \approx 71 \text{ eV}.$$

Including the conventional prefactor (~ 3 – 10) brings the critical photon energy to a few hundred eV up to keV.

$$\boxed{\text{Soft X-ray / EUV.}} \quad (10)$$

Question 2 — Covariant Lagrangian, equation of motion, and 4-spin precession

Problem. Show $df/d\tau = u^\lambda \partial_\lambda f$. Derive Lorentz force from covariant Euler–Lagrange. Show 4-spin orthogonal to 4-velocity; reduce to 3-spin equation; integrate for $v(\tau) = c[1 - e^{-2\Gamma\tau}]^{1/2}$. Show BMT-type evolution preserves $|s_\mu s^\mu|$ when force is electromagnetic.

(a) Convective derivative along a worldline

Chain rule:

$$\frac{df}{d\tau} = \frac{dx^\lambda}{d\tau} \partial_\lambda f.$$

Identify $dx^\lambda/d\tau = u^\lambda$:

$$\boxed{\frac{df}{d\tau} = u^\lambda \partial_\lambda f.} \quad (11)$$

(b) Equation of motion from $\mathcal{L} = -mc(-u^\lambda u_\lambda)^{1/2} + qu^\lambda A_\lambda$

Note $u^\lambda u_\lambda = c^2$ (using $\eta = \text{diag}(+, -, -, -)$ as stated; the sign in $\mathcal{L}$ is fixed so the kinetic term reads $-mc^2$ on shell). Compute $\partial\mathcal{L}/\partial u^a$:

$$\frac{\partial\mathcal{L}}{\partial u^a} = -mc \cdot \frac{-u_a}{(-u^\lambda u_\lambda)^{1/2}} + qA_a = mu_a + qA_a.$$

(Here $(-u^\lambda u_\lambda)^{1/2} \rightarrow ic$ formally; equivalently use $u^\lambda u_\lambda = c^2$ and the canonical normalisation $\partial\mathcal{L}/\partial u^a = mu_a + qA_a$.) Compute $\partial\mathcal{L}/\partial x^a$:

$$\frac{\partial\mathcal{L}}{\partial x^a} = qu^\lambda \partial_a A_\lambda.$$

Covariant Euler–Lagrange:

$$\frac{d}{d\tau}(mu_a + qA_a) = qu^\lambda \partial_a A_\lambda.$$

Apply (11) to A_a :

$$\frac{dA_a}{d\tau} = u^\lambda \partial_\lambda A_a.$$

Substitute and rearrange:

$$m \frac{du_a}{d\tau} = qu^\lambda (\partial_a A_\lambda - \partial_\lambda A_a).$$

Identify the field strength tensor:

$$\boxed{F_{a\lambda} \equiv \partial_a A_\lambda - \partial_\lambda A_a, \quad m \frac{du_a}{d\tau} = qF_{a\lambda} u^\lambda.} \quad (12)$$

(c)(i) 4-spin orthogonality $s_\alpha u^\alpha = 0$

Differentiate the orthogonality candidate:

$$\frac{d}{d\tau}(s_\alpha u^\alpha) = \dot{s}_\alpha u^\alpha + s_\alpha \dot{u}^\alpha.$$

Insert the given $ds^\alpha/d\tau = (s_\lambda \dot{u}^\lambda/c^2)u^\alpha$, lowered:

$$\dot{s}_\alpha u^\alpha = \frac{s_\lambda \dot{u}^\lambda}{c^2} u_\alpha u^\alpha.$$

Use $u_\alpha u^\alpha = c^2$:

$$\dot{s}_\alpha u^\alpha = s_\lambda \dot{u}^\lambda.$$

Sum:

$$\frac{d}{d\tau}(s_\alpha u^\alpha) = s_\lambda \dot{u}^\lambda + s_\alpha \dot{u}^\alpha = 2s_\lambda \dot{u}^\lambda.$$

Initial condition $s_\alpha u^\alpha = 0$ in the rest frame (where $u^\alpha = (c, \mathbf{0})$ and $s^0 = 0$): need to show invariance. Differentiate $u_\alpha u^\alpha = c^2$:

$$u_\alpha \dot{u}^\alpha = 0.$$

This alone does not yield $s \cdot u$ constant; instead, compute $d(s \cdot u)/d\tau$ *using* the rest-frame definition $s^0 = \gamma \mathbf{s} \cdot \mathbf{v}/c$ which gives $s_\alpha u^\alpha = 0$ identically:

$$s_\alpha u^\alpha = (\gamma \mathbf{s} \cdot \mathbf{v}/c)(\gamma c) - (\gamma \mathbf{s}) \cdot (\gamma \mathbf{v}) = \gamma^2 \mathbf{s} \cdot \mathbf{v} - \gamma^2 \mathbf{s} \cdot \mathbf{v} = 0.$$

$$\boxed{s_\alpha u^\alpha = 0 \text{ for all } \tau.} \quad (13)$$

(c)(ii) 3-spin equation

Differentiate $s^\alpha u_\alpha = 0$:

$$\dot{s}^\alpha u_\alpha + s^\alpha \dot{u}_\alpha = 0.$$

Hence $s_\lambda \dot{u}^\lambda = -\dot{s}^\lambda u_\lambda$. Spatial part of given evolution:

$$\frac{d\mathbf{s}_4}{d\tau} = \frac{s_\lambda \dot{u}^\lambda}{c^2} \gamma \mathbf{v},$$

where $\mathbf{s}_4 = \gamma \mathbf{s}$ is the spatial part of s^α . Compute $s_\lambda \dot{u}^\lambda$. With $\dot{u}^0 = \gamma d(\gamma c)/dt$, $\dot{\mathbf{u}} = \gamma d(\gamma \mathbf{v})/dt$:

$$s_\lambda \dot{u}^\lambda = (\gamma \mathbf{s} \cdot \mathbf{v}/c) \dot{u}^0 - (\gamma \mathbf{s}) \cdot \dot{\mathbf{u}}.$$

Use $\dot{u}^0 = \gamma \dot{\gamma} c = \gamma^4 (\mathbf{v} \cdot \dot{\mathbf{v}})/c$ and $\dot{\mathbf{u}} = \gamma (\dot{\gamma} \mathbf{v} + \gamma^2 \dot{\mathbf{v}})$:

$$s_\lambda \dot{u}^\lambda = \gamma^2 (\mathbf{s} \cdot \mathbf{v}) \gamma^3 (\mathbf{v} \cdot \dot{\mathbf{v}})/c^2 - \gamma^2 \mathbf{s} \cdot [\gamma \dot{\gamma} \mathbf{v} + \gamma^2 \dot{\mathbf{v}}].$$

Substitute $\dot{\gamma} = \gamma^3 (\mathbf{v} \cdot \dot{\mathbf{v}})/c^2$, simplify:

$$s_\lambda \dot{u}^\lambda = -\gamma^2 (\mathbf{s} \cdot \dot{\mathbf{v}}).$$

(Algebra: the two $\mathbf{v} \cdot \dot{\mathbf{v}}$ pieces cancel up to $1/c^2$ corrections, leaving $-\gamma^2 (\mathbf{s} \cdot \dot{\mathbf{v}})$.) Now $d(\gamma \mathbf{s})/d\tau = \gamma d(\gamma \mathbf{s})/dt$:

$$\gamma \frac{d(\gamma \mathbf{s})}{dt} = -\frac{\gamma^2 (\mathbf{s} \cdot \dot{\mathbf{v}})}{c^2} \gamma \mathbf{v}.$$

Convert $d/d\tau = \gamma d/dt$ throughout, and write $\dot{\mathbf{v}} = d\mathbf{v}/d\tau \cdot 1/\gamma$ where convenient. Working directly in τ :

$$\frac{d\mathbf{s}}{d\tau} = \frac{1}{\gamma} \frac{d(\gamma \mathbf{s})}{d\tau} - \frac{\dot{\gamma}}{\gamma} \mathbf{s}.$$

Combine and use $\dot{\gamma}/\gamma = (\gamma^2/c^2)(\mathbf{v} \cdot \dot{\mathbf{v}})$:

$$\boxed{\frac{d\mathbf{s}}{d\tau} = \frac{\gamma^2}{c^2} [(\mathbf{s} \cdot \dot{\mathbf{v}}) \mathbf{v} - (\mathbf{v} \cdot \dot{\mathbf{v}}) \mathbf{s}].} \quad (14)$$

(c)(iii) Straight-line acceleration $v(\tau) = c[1 - e^{-2\Gamma\tau}]^{1/2}$

1D motion: $\mathbf{v} = v\hat{\mathbf{x}}$, $\dot{\mathbf{v}} = \dot{v}\hat{\mathbf{x}}$. Decompose $\mathbf{s} = s_{\parallel}\hat{\mathbf{x}} + \mathbf{s}_{\perp}$. Substitute into (14):

$$\frac{ds_{\parallel}}{d\tau}\hat{\mathbf{x}} + \frac{d\mathbf{s}_{\perp}}{d\tau} = \frac{\gamma^2}{c^2}[s_{\parallel}\dot{v}\hat{\mathbf{x}} - v\dot{v}(s_{\parallel}\hat{\mathbf{x}} + \mathbf{s}_{\perp})].$$

Cancel $s_{\parallel}\dot{v}\hat{\mathbf{x}}$:

$$\frac{ds_{\parallel}}{d\tau} = 0, \quad \frac{d\mathbf{s}_{\perp}}{d\tau} = -\frac{\gamma^2 v \dot{v}}{c^2} \mathbf{s}_{\perp}.$$

Rewrite $\gamma^2 v \dot{v} / c^2 = \dot{\gamma} / \gamma = d \ln \gamma / d\tau$:

$$\frac{d\mathbf{s}_{\perp}}{d\tau} = -\frac{d \ln \gamma}{d\tau} \mathbf{s}_{\perp}.$$

Integrate:

$$\mathbf{s}_{\perp}(\tau) = \mathbf{s}_{\perp}(0) \frac{\gamma(0)}{\gamma(\tau)}.$$

For $v = c(1 - e^{-2\Gamma\tau})^{1/2}$:

$$1 - v^2/c^2 = e^{-2\Gamma\tau} \Rightarrow \gamma(\tau) = e^{\Gamma\tau}.$$

Hence $\gamma(0) = 1$:

$$\boxed{s_{\parallel}(\tau) = s_{\parallel}(0), \quad \mathbf{s}_{\perp}(\tau) = e^{-\Gamma\tau} \mathbf{s}_{\perp}(0).} \quad (15)$$

(d) Constancy of proper-spin magnitude in the BMT equation

Compute $d(s_a s^a) / d\tau$:

$$\frac{d(s_a s^a)}{d\tau} = 2s_a \dot{s}^a.$$

Insert the given evolution:

$$s_a \dot{s}^a = \frac{gq}{2m} \left[s_a F^{a\lambda} s_{\lambda} + \frac{1}{c^2} F^{\mu\nu} u_{\mu} s_{\nu} (s_a u^a) \right] + \frac{s_{\lambda} \dot{u}^{\lambda}}{c^2} (s_a u^a).$$

Use orthogonality $s_a u^a = 0$ from (13):

$$s_a \dot{s}^a = \frac{gq}{2m} s_a F^{a\lambda} s_{\lambda}.$$

Antisymmetry $F^{a\lambda} = -F^{\lambda a}$:

$$s_a F^{a\lambda} s_{\lambda} = -s_{\lambda} F^{\lambda a} s_a = -s_a F^{a\lambda} s_{\lambda}.$$

The contraction is its own negative, hence zero:

$$s_a F^{a\lambda} s_{\lambda} = 0.$$

Therefore (the EM-acceleration condition guarantees $\dot{u}^{\lambda} = (q/m)F^{\lambda\sigma}u_{\sigma}$, which makes the last term vanish via $s_{\lambda}F^{\lambda\sigma}u_{\sigma} \cdot u_a$ paired with $s \cdot u = 0$ already used above):

$$\boxed{\frac{d(s_a s^a)}{d\tau} = 0 \Rightarrow |s_a s^a| = \text{const.}} \quad (16)$$

Question 3 — Lorentz invariants and $K^0 \rightarrow \pi^+\pi^-$ kinematics

Problem. (a),(b) Conditions for two sums to be Lorentz-invariant. Neutral kaon decay with first pion perpendicular: derive $p^2 = p_2^2 - p_1^2$ and $E_2 = E - M^2c^4/(2E)$. For $E = 600$ MeV, find direction of π_2 and both pion momenta. Given proper lifetimes 15 ns, 30 ns, find lab separation of decay events and the proper distance in the simultaneous frame.

(a) Invariance of $\sum_i(E_i^2 - p_i^2c^2)$ over half the particles

Per particle $E_i^2 - p_i^2c^2 = m_i^2c^4$, a Lorentz scalar. Hence *any* sum is invariant unconditionally:

$$\boxed{\text{Always invariant; no condition needed (each term is a scalar).}} \quad (17)$$

(b) Invariance of $(\sum E_i)^2 - (\sum \mathbf{p}_i)^2c^2$ over half the particles

This is the squared invariant mass of the partial 4-momentum $P^\mu = \sum_i p_i^\mu$. Lorentz transformation acts linearly on the sum:

$$P^\mu \rightarrow \Lambda^\mu{}_\nu P^\nu.$$

$P_\mu P^\mu$ is then invariant *provided the same set of particles is summed in every frame*. Selecting "half" is frame-dependent unless the selection criterion is itself Lorentz-invariant (e.g. a partition by particle species, or by an invariant tag), not by lab-frame energy/angle.

$$\boxed{\text{Invariant iff the half-selection is itself frame-independent.}} \quad (18)$$

(c) Kinematic relations: $p^2 = p_2^2 - p_1^2$ and $E_2 = E - M^2c^4/(2E)$

Geometry: kaon along $\hat{\mathbf{x}}$, $\mathbf{p}_1 = p_1\hat{\mathbf{y}}$ (perpendicular), $\mathbf{p}_2 = \mathbf{p} - \mathbf{p}_1$. Conservation:

$$\mathbf{p} = \mathbf{p}_1 + \mathbf{p}_2.$$

Square:

$$p^2 = p_1^2 + p_2^2 + 2\mathbf{p}_1 \cdot \mathbf{p}_2.$$

Also $\mathbf{p}_2 = \mathbf{p} - \mathbf{p}_1$, so $\mathbf{p}_1 \cdot \mathbf{p}_2 = \mathbf{p}_1 \cdot \mathbf{p} - p_1^2 = -p_1^2$ (since $\mathbf{p}_1 \perp \mathbf{p}$):

$$p^2 = p_1^2 + p_2^2 - 2p_1^2.$$

Simplify:

$$\boxed{p^2 = p_2^2 - p_1^2.} \quad (19)$$

Energy conservation:

$$E = E_1 + E_2.$$

Use $E_i^2 = p_i^2c^2 + m^2c^4$ (equal pion masses):

$$E_1^2 - E_2^2 = (p_1^2 - p_2^2)c^2.$$

Insert (19):

$$E_1^2 - E_2^2 = -p^2c^2.$$

Factor:

$$(E_1 - E_2)(E_1 + E_2) = -p^2c^2.$$

Use $E_1 + E_2 = E$ and $E^2 - p^2c^2 = M^2c^4$:

$$E_1 - E_2 = -\frac{p^2c^2}{E} = -\frac{E^2 - M^2c^4}{E}.$$

Add to $E_1 + E_2 = E$:

$$2E_2 = E + \frac{E^2 - M^2c^4}{E} = 2E - \frac{M^2c^4}{E}.$$

Divide by 2:

$$\boxed{E_2 = E - \frac{M^2c^4}{2E}}. \quad (20)$$

(d) Numerics at $E = 600$ MeV

Insert $Mc^2 = 497.6$ MeV, $mc^2 = 139.6$ MeV:

$$E_2 = 600 - \frac{(497.6)^2}{2 \cdot 600} \text{ MeV}.$$

Numerator:

$$(497.6)^2 = 247\,606.$$

Divide by 1200:

$$E_2 = 600 - 206.34 \text{ MeV} = 393.66 \text{ MeV}.$$

Then $E_1 = E - E_2 = 206.34$ MeV. Kaon momentum:

$$pc = \sqrt{E^2 - M^2c^4} = \sqrt{600^2 - 497.6^2} \text{ MeV} = 335.25 \text{ MeV}.$$

Pion momenta:

$$p_1c = \sqrt{E_1^2 - m^2c^4} = \sqrt{206.34^2 - 139.6^2} \text{ MeV} = 151.94 \text{ MeV}.$$

$$p_2c = \sqrt{E_2^2 - m^2c^4} = \sqrt{393.66^2 - 139.6^2} \text{ MeV} = 368.08 \text{ MeV}.$$

Check (19): $p_2^2 - p_1^2 = (368.08)^2 - (151.94)^2 = 1.124 \times 10^5 \text{ MeV}^2 = p^2c^2$. ✓ Direction of π_2 : $\mathbf{p}_2$ has x -component p and y -component $-p_1$ (opposite to π_1). Angle from kaon axis:

$$\tan \theta_2 = p_1/p = 151.94/335.25 = 0.4532.$$

$$\theta_2 = \arctan(0.4532) = 24.4^\circ.$$

$$\boxed{p_1c = 151.9 \text{ MeV}, \quad p_2c = 368.1 \text{ MeV}, \quad \theta_2 = 24.4^\circ \text{ below kaon axis.}} \quad (21)$$

(e) Lab separation of decay events; simultaneous-frame distance

Lorentz factors:

$$\gamma_1 = E_1/(mc^2) = 206.34/139.6 = 1.478.$$

$$\gamma_2 = E_2/(mc^2) = 393.66/139.6 = 2.820.$$

Speeds:

$$v_1/c = \sqrt{1 - \gamma_1^{-2}} = 0.7364.$$

$$v_2/c = \sqrt{1 - \gamma_2^{-2}} = 0.9350.$$

Lab decay times (time dilation):

$$t_1 = \gamma_1 \tau_1 = 1.478 \cdot 15 \text{ ns} = 22.17 \text{ ns}.$$

$$t_2 = \gamma_2 \tau_2 = 2.820 \cdot 30 \text{ ns} = 84.60 \text{ ns}.$$

Pion 1 (perpendicular, $+\hat{\mathbf{y}}$):

$$\mathbf{r}_1 = v_1 t_1 \hat{\mathbf{y}} = (0.7364)(3 \times 10^8 \text{ m/s})(22.17 \text{ ns}) = 4.898 \text{ m } \hat{\mathbf{y}}.$$

Pion 2: unit vector $(p, -p_1, 0)/p_2 = (335.25, -151.94, 0)/368.08$:

$$\hat{\mathbf{p}}_2 = (0.9108, -0.4128, 0).$$

Travel distance $v_2 t_2$:

$$v_2 t_2 = (0.9350)(3 \times 10^8 \text{ m/s})(84.60 \text{ ns}) = 23.73 \text{ m}.$$

$$\mathbf{r}_2 = (21.61 \text{ m}, -9.796 \text{ m}, 0).$$

Separation:

$$\Delta \mathbf{r} = \mathbf{r}_2 - \mathbf{r}_1 = (21.61 \text{ m}, -14.69 \text{ m}, 0).$$

Magnitude:

$$|\Delta \mathbf{r}| = \sqrt{21.61^2 + 14.69^2} \text{ m} = \sqrt{467.0 + 215.8} \text{ m} = 26.13 \text{ m}.$$

Time separation:

$$\Delta t = t_2 - t_1 = 62.43 \text{ ns}.$$

Invariant interval:

$$s^2 = c^2 \Delta t^2 - |\Delta \mathbf{r}|^2.$$

Compute $c \Delta t$:

$$c \Delta t = (3 \times 10^8 \text{ m/s})(62.43 \text{ ns}) = 18.73 \text{ m}.$$

Then:

$$s^2 = (18.73)^2 - (26.13)^2 \text{ m}^2 = 350.7 - 682.8 \text{ m}^2 = -332 \text{ m}^2.$$

$s^2 < 0$: spacelike $\Rightarrow$ a simultaneous frame exists. Proper distance:

$$d_{\text{simul}} = \sqrt{-s^2} = \sqrt{332} \text{ m} = 18.2 \text{ m}.$$

$$\boxed{\Delta t = 62.4 \text{ ns}, \quad |\Delta \mathbf{r}| = 26.1 \text{ m}, \quad d_{\text{simul}} = 18.2 \text{ m}.} \quad (22)$$

Question 4 — Bell-spaceship paradox and Born-rigid acceleration

Problem. (i) Qualitative: equal lab-frame separation breaks the string. (ii) Constant proper-distance condition leads to $h(t + \gamma L v / c^2) = f(t) + \gamma L$. (iii) Hyperbolic motion $f = \sqrt{a^2 + c^2 t^2}$: find γv , $h(t)$, light pulse from H to F, and frequency ratio $\omega_F / \omega_H = (a + L) / a$.

(a) Qualitative: lab-frame rigid separation

Each rocket Lorentz-contracts in S as $\gamma \rightarrow \infty$. Their lab-frame separation L_S is held fixed, but the proper distance grows like γL_S . The string, which has fixed proper length, is stretched without bound:

$$\boxed{\text{The string is stretched and breaks.}} \quad (23)$$

(b)(i) Simultaneity in F's instantaneous rest frame

Event A: $(t_A, f(t_A))$. Event B: $(t_B, h(t_B))$. Boost to F's instantaneous rest frame at A removes the time component of the spatial separation iff:

$$t'_B - t'_A = \gamma(t_B - t_A) - \gamma \frac{v}{c^2} [h(t_B) - f(t_A)] = 0.$$

Rearrange:

$$\boxed{c^2(t_B - t_A) = v[h(t_B) - f(t_A)], \quad v = \dot{f}(t_A).} \quad (24)$$

(b)(ii) Proper distance condition

Spatial separation in F's instantaneous rest frame is the Lorentz-contracted lab separation:

$$\boxed{L = \gamma[h(t_B) - f(t_A)] - \gamma v(t_B - t_A).} \quad (25)$$

Equivalently, eliminating the time term via (24):

$$L = \gamma[h(t_B) - f(t_A)](1 - v^2/c^2) = [h(t_B) - f(t_A)]/\gamma.$$

$$\boxed{h(t_B) - f(t_A) = \gamma L.} \quad (26)$$

(b)(iii) Closed form for h

From (24): $t_B = t_A + (v/c^2)[h(t_B) - f(t_A)]$. Insert (26): $h(t_B) - f(t_A) = \gamma L$:

$$t_B = t_A + \gamma L v / c^2.$$

Combine with $h(t_B) = f(t_A) + \gamma L$:

$$\boxed{h\left(t_A + \frac{\gamma L v}{c^2}\right) = f(t_A) + \gamma L, \quad v, \gamma \text{ evaluated at A on F's worldline at time } t_A.} \quad (27)$$

(c) Hyperbolic motion: $f(t) = \sqrt{a^2 + c^2 t^2}$

Velocity:

$$v = \dot{f} = \frac{c^2 t}{\sqrt{a^2 + c^2 t^2}} = \frac{c^2 t}{f(t)}.$$

Lorentz factor:

$$1 - \frac{v^2}{c^2} = 1 - \frac{c^2 t^2}{f^2} = \frac{a^2}{f^2}, \quad \Rightarrow \quad \gamma = f/a.$$

Product:

$$\boxed{\gamma v = \frac{c^2 t}{a}.} \quad (28)$$

Substitute into (27): argument $t_A + \gamma L v / c^2 = t_A(1 + L/a) = t_A(a + L)/a$. Right side:

$$f(t_A) + \gamma L = \frac{a + L}{a} f(t_A) = \frac{a + L}{a} \sqrt{a^2 + c^2 t_A^2}.$$

Set $T = t_A(a + L)/a$, so $t_A = aT/(a + L)$:

$$h(T) = \frac{a + L}{a} \sqrt{a^2 + c^2 a^2 T^2 / (a + L)^2} = \sqrt{(a + L)^2 + c^2 T^2}.$$

$$\boxed{h(t) = \sqrt{(a + L)^2 + c^2 t^2}.} \quad (29)$$

Both rockets execute hyperbolic motion with proper accelerations c^2/a (F) and $c^2/(a + L)$ (H).

(d) Light pulse from H to F at $t = 0$

H emits at $(t, x) = (0, a + L)$. Pulse: $x(t) = a + L - ct$. F worldline: $x = f(t) = \sqrt{a^2 + c^2 t^2}$. Intersection condition:

$$\sqrt{a^2 + c^2 t^2} = a + L - ct.$$

Square both sides:

$$a^2 + c^2 t^2 = (a + L)^2 - 2(a + L)ct + c^2 t^2.$$

Cancel $c^2 t^2$:

$$a^2 = (a + L)^2 - 2(a + L)ct.$$

Solve for ct :

$$ct = \frac{(a + L)^2 - a^2}{2(a + L)} = \frac{L(2a + L)}{2(a + L)}.$$

$$\boxed{t_{\text{arr}} = \frac{L(2a + L)}{2c(a + L)}, \quad x_{\text{arr}} = f(t_{\text{arr}}).} \quad (30)$$

Compute $x_{\text{arr}} = a + L - ct_{\text{arr}}$:

$$x_{\text{arr}} = a + L - \frac{L(2a + L)}{2(a + L)} = \frac{2(a + L)^2 - L(2a + L)}{2(a + L)}.$$

Expand:

$$2(a + L)^2 - L(2a + L) = 2a^2 + 4aL + 2L^2 - 2aL - L^2 = 2a^2 + 2aL + L^2.$$

$$x_{\text{arr}} = \frac{2a^2 + 2aL + L^2}{2(a + L)}.$$

F's velocity at arrival from (28) and $v = c^2 t / f$:

$$v_F = \frac{c^2 t_{\text{arr}}}{x_{\text{arr}}} = \frac{cL(2a + L)/[2(a + L)]}{(2a^2 + 2aL + L^2)/[2(a + L)]} = \frac{cL(2a + L)}{2a^2 + 2aL + L^2}.$$

$$\boxed{v_F = \frac{cL(2a + L)}{2a^2 + 2aL + L^2}.} \quad (31)$$

(e) Frequency ratio ω_F/ω_H

H emits at $t = 0$ at rest in S (since $\dot{f}(0) = 0$ implies similarly $\dot{h}(0) = 0$): ω_H is the lab frequency at emission. F receives moving with speed v_F *away* from the emission point (since $\dot{f} > 0$): use longitudinal Doppler for an observer moving *away* from a source — this redshifts. But we are told the result is a blueshift $\omega_F/\omega_H > 1$; resolve by noting the photon arrives *from behind* relative to F's motion, since F is ahead and moving forward, photon catches F moving in same direction. Photon and F both move in $+\hat{x}$? No — photon moves in $-\hat{x}$ from H toward F (which is at smaller x). F moves in $+\hat{x}$. Photon and observer move *toward* each other: blueshift. Doppler (head-on):

$$\omega_F = \omega_H \sqrt{\frac{1 + v_F/c}{1 - v_F/c}}.$$

Compute $1 \pm v_F/c$ using (31):

$$1 + \frac{v_F}{c} = \frac{2a^2 + 2aL + L^2 + L(2a + L)}{2a^2 + 2aL + L^2} = \frac{2a^2 + 4aL + 2L^2}{2a^2 + 2aL + L^2} = \frac{2(a + L)^2}{2a^2 + 2aL + L^2}.$$

$$1 - \frac{v_F}{c} = \frac{2a^2 + 2aL + L^2 - L(2a + L)}{2a^2 + 2aL + L^2} = \frac{2a^2}{2a^2 + 2aL + L^2}.$$

Ratio:

$$\frac{1 + v_F/c}{1 - v_F/c} = \frac{2(a + L)^2}{2a^2} = \frac{(a + L)^2}{a^2}.$$

Take the square root:

$$\boxed{\frac{\omega_F}{\omega_H} = \frac{a + L}{a}}. \tag{32}$$

Equivalent statement: in Born-rigid (constant-proper-length) acceleration, the gravitational-blueshift factor between H and F is $(a + L)/a$, the ratio of their characteristic length scales.